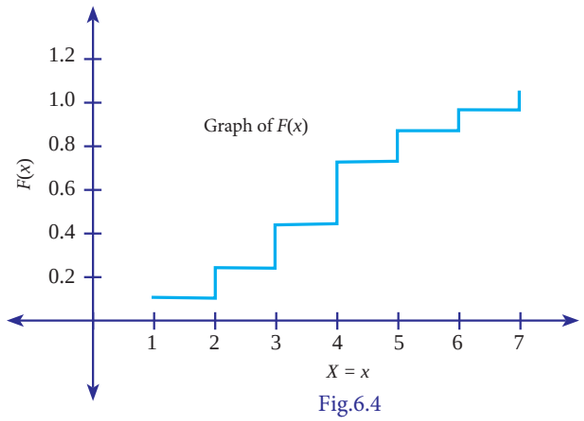


Fig.6.4

$$F(x) \text{ is } F_X(x) = \begin{cases} 0, & if\ x<1 \\ 0.10, & if\ x\le 1 \\ 0.22, & if\ x\le 2 \\ 0.42, & if\ x\le 3 \\ 0.72, & if\ x\le 4 \\ 0.87, & if\ x\le 5 \\ 0.95, & if\ x\le 6 \\ 1, & if\ x\le 7 \end{cases}$$

6.1.3 Continuous random variable

Definition 6.5

A random variable X which can take on any value (integral as well as fraction) in the interval is called continuous random variable.

Examples of continuous random variable

- The amount of water in a 10 ounce bottle.
- The speed of a car.
- Electricity consumption in kilowatt hours.
- Height of people in a population.
- Weight of students in a class.
- The length of time taken by a truck driver to go from Chennai to Madurai, etc.

Probability density function

Definition 6.6

The probability that a random variable X takes a value in the interval $[t_1, t_2]$ (open or closed) is given by the integral of a function called the probability density function $f_X(x)$:

$$P(t_1 \le X \le t_2) = \int_{t_1}^{t_2} f_X(x)dx .$$

Other names that are used instead of probability density function include density function, continuous probability function, integrating density function.

The probability density functions $f_X(x)$ or simply by $f(x)$ must satisfy the following conditions.

(i) $f(x) \ge 0\ \forall\ x$ and (ii) $\int_{-\infty}^{\infty} f(x)dx = 1$.

Example 6.8

A continuous random variable X has the following p.d.f $f(x) = ax\ ,\ 0 \le x \le 1$

Determine the constant a and also find $P\left[X \le \frac{1}{2}\right]$

Solution:

We know that

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

$$\int_0^1 ax\ dx = 1 \Rightarrow a\int_0^1 xdx = 1$$

$$\Rightarrow a\left(\frac{x^2}{2}\right)_0^1 = 1$$

$$\Rightarrow \frac{a}{2}(1-0) = 1$$

$$\Rightarrow a = 2$$

$$P\left[x \le \frac{1}{2}\right] = \int_{-\infty}^{\frac{1}{2}} f(x)dx$$

$$= \int_0^{\frac{1}{2}} axdx \quad = \int_0^{\frac{1}{2}} 2xdx \quad = \frac{1}{4} \backslash$$

Example 6.9

A continuous random variable X has p.d.f $f(x) = 5x^4, 0 \le x \le 1$ Find a_1 and a_2 such that (i) $P[X \le a_1] = P[X > a_1]$ (ii) $P[X > a_2] = 0.05$

Solution

(i) Since $P[X \le a_1] = P[X > a_1]$

$$P[X \le a_1] = \frac{1}{2}$$

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